MATH 3- QUIZ # 1

Total points 17/25

Email *

acearthur.egno@cvsu.edu.ph

Students Information

0 of 0 points

Student Number *

202311301

Section *

BSCS 3-5

First name *

Ace Arthur

Middle name *

Balante

Last name *

Egno

Read each item carefully. Choose and tick the letter that corresponds with the best answer.

16 of 20 points

The value of a determinant helps determine whether a matrix is_____. * 1/1

- A. scalar
- B. diagonal
- C. symmetric
- D. singular or non-singular

For which type of matrix is the determinant defined? * 1/1

- A. Diagonal
- B. Rectangular
- C. Square
- D. Trinagular

What do you call the matrix if all pivots are 1 and each pivot column contains only one nonzero entry? *0/1

- A. Diagonal form
- B. Inverse form
- C. Reduced Echelon Form
- D. Row echelon form but nor reduced echelon form

The purpose of Gaussian elimination is to transform the matrix into____. * 1/1

- A. zero form
- B. symmetric form
- C. lower triangular form
- D. upper triangular form

The inverse method can be applied if the determinants is _____. * 1/1

- A. zero
- B. nonzero
- C. negative
- D. 1

How are the rows of the coefficient matrix related when a system of equations has infinitely many solutions? *1/1

- A. the coefficients are zero.
- B. the coefficients are identical.
- C. the coefficients are dependent
- D. the coefficients are independent

Which row operation changes the sign of the determinant? * 1/1

- (●) A. Interchanging two rows
- ( ) B. Multiplying a row by zero
- ( ) C. Multiplying a row by nonzero scalar
- ( ) D. Adding a scalar multiple of one row to another.

The matrix form of a linear system is written as_____. * 1/1

- ( ) A. A + X = B
- (●) B. AX = B
- ( ) C. AB = X
- ( ) D. A = XB

A determinant is used to find the _____ of a matrix * 1/1

- ( ) A. adjugate
- (●) B. inverse
- ( ) C. transpose
- ( ) D. triangular

REF is primarily used to simplify solving_____. * 1/1

- A. determinants only
- B. inverse matrices only
- C. quadratics equations
- D. systems of linear equations

The process of obtaining RREF is an extension of _____ * 1/1

- A. Matrix addition
- B. Cofactor expansion
- C. Inverse multiplication
- D. Gaussian elimination

The determinant of the identity matrix is_____. * 1/1

- A. -1
- B. 0
- C. 1
- D. 2

If two identical row exists in a matrix, then the determinant is ___. * 1/1

- A. negative
- B. positive
- C. zero
- D. Either positive or negative

The cofactor of an element a sub ij is defined as____. * 1/1

- A. the determinant itself
- B. the product of all elements.
- C. its minor with a negative sign
- D. its minor multiplied by (-1)^i+j

If IAI = 5, what is I2AI for a 3 x 3 matrix? * 0/1

- A. 10
- B. 20
- C. 30
- D. 40

If IAI =3, find inverse of A, A^-1 * 0/1

- (●) A. 1/3
- ( ) B. 1/9
- ( ) C. 3
- ( ) D. 9

In REF, all entries below a pivot must be ___) * 1/1

- ( ) A. -1
- (●) B. 0
- ( ) C. 1
- ( ) D. either 0 and 1

What is the sum of determinant A and B? * 1/1

$A=\begin{bmatrix}3 & -6\\0 & 3\end{bmatrix}$ $B=\begin{bmatrix}2 & -4\\-2 & 5\end{bmatrix}$

- ( ) A. 9
- ( ) B. 10
- (●) C. 11
- ( ) D. 12

The Gaussian elimination method converts the augmented matrix into_____ * 0/1

- A. Diagonal form
- B. Identity matrix
- C. Triangular form
- D. Reduced row echelon form

What is the main purpose of using matrices in solving systems of equations? * 1/1

- A. remove constants
- B. avoid using arithmetic
- C. simplify large systems
- D. make equations more complex

II. **True or False.** Write **TRUE** if the statement is correct. If it is false, write the correct answer. 1 of 5 points

REF is used to find the inverse of a matrix * ···/1

FALSE

RREF is used to simplify complex equations. * 1/1

TRUE

Gaussian elimination produces RREF directly. * ···/1

FALSE

If the determinant of the coefficient matrix is zero, the system has a unique solution. *···/1

FALSE

A linear system can have one solution, no solution, or infinitely many solutions. *···/1

TRUE

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